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POSTER

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Published: 14 July 2025

[Citation in BibTeX format](#)

GECCO '25 Companion: Genetic and
Evolutionary Computation Conference
Companion
July 14 - 18, 2025
Malaga, Spain

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An LLM-Based Genetic Algorithm for Prompt Engineering

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Abstract

Large Language Models (LLMs) have gained recognition as valuable assets across virtually all industries, yet they rely heavily on manually crafted input prompts. In real-world applications, the dependence on specialized staff, skilled prompt engineering, and domain-specific knowledge often leads to suboptimal performance and increased costs. This paper investigates the use of Genetic Algorithms (GAs) to autonomously generate, evolve, and judge LLM prompts. Specifically, we customized a standard GA implementation to handle textual individuals, which are manipulated by LLM-guided genetic operators that iteratively create and enhance candidate prompts. Additionally, LLMs are employed to assess the correctness of outputs, forming the basis of our GA's fitness function. Our findings suggest that GA-driven prompt engineering can consistently produce solutions that are superior in both accuracy and efficiency compared to those acquired by prompt engineers who possess technical skills but lack domain-specific knowledge and a full understanding of vendor-specific prompting idiosyncrasies. This conclusion is supported by experimental results obtained from four public datasets and three modern LLMs developed by OpenAI, Meta, and Mistral AI. Ultimately, this study highlights the viability and potential of fully automated optimization tools in minimizing human effort in writing high-performance prompts.

Keywords

Genetic Algorithm, LLM prompt optimization, prompt engineering.

ACM Reference Format:

Leandro A. Loss and Pratikkumar Dhuvad. 2025. An LLM-Based Genetic Algorithm for Prompt Engineering. In *Genetic and Evolutionary Computation Conference (GECCO '25 Companion)*, July 14–18, 2025, Malaga, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3712255.3726633>

1 Introduction

Large Language Models (LLMs) have significantly influenced the field of Artificial Intelligence (AI). By capitalizing on transformer-based neural networks, extensive datasets, and reinforcement learning approaches, these models exhibit a strong capacity for understanding and generating human-like text. Nevertheless, their accuracy still relies heavily on the clarity and specificity of user-provided instructions, a process known as *prompt engineering*. This reliance

poses considerable challenges for teams operating with limited resources or handling domain-specific content. Consequently, automating prompt engineering has the potential to substantially enhance LLM performance across a wide range of tasks and industries by reducing dependence on specialized input.

The primary goal of this work is to evaluate the feasibility of employing Genetic Algorithms (GAs) to automate prompt engineering in LLMs. By treating prompts as individuals in a GA, one can systematically explore numerous configurations that may surpass manual methods. Specifically, we introduce modifications to standard GA operators to handle textual individuals, data samples, and *meta prompts* (i.e., generic LLM instructions), guiding the GA in iteratively generating and refining prompts and thereby promoting their optimization. To test the viability of this approach, we conduct experiments to address the following questions: (i) Can GAs reliably discover high-performing prompts without human intervention or domain-specific expertise? (ii) How do GA-derived prompts compare to those developed by human experts? Our empirical findings are encouraging and suggest that GA-optimized prompts are both quantitatively and qualitatively superior to those obtained through manual crafting.

2 Related Literature

Despite their notable strengths, the adoption of LLMs remain difficult in areas with limited technical expertise. Manual prompt engineering continues to be a key element for optimizing LLM performance [13], but this requirement can deter non-experts from fully leveraging these models. In response, researchers have proposed various semi-automated approaches to simplify LLM usage. Representative efforts include Shin et al.'s AutoPrompt [21], which employs gradient-guided search for prompt creation, and FLUENT PROMPT by Shi et al. [20], which incorporates linguistic constraints to prompt generation. Reinforcement learning-based frameworks, such as RLPrompt [6] and TEMPERA [25], further introduce dynamic methods for adapting prompts. Other tools, including MetaPrompting [10], Automatic Prompt Optimization [18], and Learning-to-Plan [8], further reduce the reliance on expert knowledge, extending LLM accessibility across various domains.

Within the broader context of optimization, GAs have long been recognized as effective solutions for complex tasks, demonstrating versatility in fields ranging from engineering design to financial modeling, owing to their ability to explore large solution spaces. Although the application of GAs in NLP is still emerging, it is gaining momentum. For example, Lehman et al. [12] and Meyerson et al. [16] illustrate the potential of LLMs to enhance genetic programming and prompt evolution, while Tanaka et al. [22] introduce a GA-based approach to prompt engineering specifically for Japanese. Additional initiatives, such as EvoPrompting [4], integrate

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GECCO '25 Companion, July 14–18, 2025, Malaga, Spain

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ACM ISBN 979-8-4007-1464-1/2025/07

<https://doi.org/10.1145/3712255.3726633>

LLMs in evolutionary searches for optimal neural architectures, whereas Guo et al. [7] and Hsieh et al. [11] explore GA-based prompt optimization for diverse tasks. Although these developments are encouraging, they often require human guidance in the form of constrained initial criteria or domain-specific input. Nonetheless, ongoing research continues to explore the role of LLMs as evolutionary operators in GAs, aiming to mitigate these bottlenecks.

3 Proposed Method

An LLM-Based Genetic Algorithm: In this study, we adapted a standard GA to handle textual individuals, which are manipulated through operators defined by LLMs. Each individual in our approach is represented as an LLM prompt, serving as a potential solution to the given task. We substituted traditional genetic operators – initialization, crossover, and mutation – with meta prompts, which are domain-agnostic prompts designed to facilitate prompt creation and the evolution of individuals. The idea of domain-agnostic meta prompts has also been investigated by Yang et al. [23], although Ye et al. [24] have critiqued their efficacy. Our method distinguishes itself by minimizing user interventions and avoiding extensive domain-specific guidelines, instead leveraging insights inferred directly from data samples. To sustain diversity and mitigate early convergence, we employed a tournament selection mechanism of size 3. Furthermore, to ensure consistency across experiments and to manage computational costs, we set the population size to 20 individuals and limited the number of generations to 20. Based on preliminary experiments, this configuration was sufficient for most tasks and datasets. Fitness was evaluated using 500 randomly pre-selected samples, even when larger datasets were available, thereby enabling us to gauge efficiency under reproducible and controlled conditions. The complete method is presented in Algorithm I, and all corresponding code and datasets are released under Apache-2.0 licensing in [15].

Algorithm I: LLM-based GA for Prompt Optimization

- 0: **input:** population size, number of generations, mutation probability, *LLM name*, and *data samples*
- 1: Initialize population using *system* and *initialization meta prompts*, and *data samples*
- 2: Evaluate fitness of individual prompts using *fitness* and *system fitness meta prompts*
- 3: While number of generations is not reached:
- 4: Perform selection using trio tournament selection
- 5: Perform pair crossover using *crossover meta prompt*
- 6: Perform conditional mutation using *mutation meta prompt*
- 7: Evaluate fitness of resulting prompts using *fitness* and *system fitness meta prompts*
- 8: Increment generation count
- 9: End While loop
- 10: **output:** individual prompt with highest fitness

Meta Prompts: Our approach refines the GA by substituting its traditional random initialization and bitwise operators with textual instructions. These instructions, referred to here as meta prompts, are predefined and domain agnostic prompts that guide the language model in generating, modifying, and evaluating individuals,

also represented as prompts, thereby simulating processes such as initialization, crossover, mutation, and fitness assessment. Our meta prompts (Table 1) have been tailored to utilize only generic language, sample data (for initialization), prompts previously produced by GA (for crossover and mutation), and corresponding output predictions (for fitness evaluation). This design choice enables our method to be readily adapted to a wide range of supervised learning tasks framed by input-output relationships.

More specifically, at the LLM-based initialization stage, a meta prompt directs the GA to create a population of candidate prompts that each address the target problem in a unique way, based on provided sample data. Meanwhile, mutation and crossover meta prompts further instruct the GA to derive new prompts using one or two existing prompts, respectively. Lastly, another meta prompt instructs the GA's fitness function to compare the model-generated outputs with ground truth for each candidate prompt. To enforce consistency and reliability in the LLM's responses, system prompts were designed to provide fundamental guardrails and shape the model's overall behavior.

Table 1: Meta prompts

Role	Prompt
system	You are an AI that helps people solve problems. Avoid comments outside the proposed prompt as the user will use your answer to integrate with another downstream system.
initial.	Create a [LLM_MODEL] prompt that solves the problem exemplified by the following examples: [SAMPLE_QUESTION_1] [SAMPLE_ANSWER_1] [SAMPLE_QUESTION_1] [SAMPLE_ANSWER_2] ...
crossover	Given the following drafts for two prompts that aim to solve the same particular problem, create a better prompt using only ideas from them. [PROMPT_A] [PROMPT_B]
mutation	Given the following draft for a prompt that aims to solve a particular problem, create a better prompt using ideas from it. [PROMPT_A]
system fitness	You are an AI that validates automated answers against ground truth. You only answer 'yes' or 'no' with no extra comments, notes, or explanations as the user will use your answer to integrate with another downstream system.
fitness	My ground truth is [TRUTH]. Does the automated output [ANSWER] linguistically, symbolically, conceptually, or fundamentally match my ground truth?

Fitness Function: The design of the fitness function plays a critical role in the effectiveness and efficiency of GAs. In an effort to preserve the automated nature of the optimization process, we introduced meta prompts to guide the LLM in assessing and comparing the ground truth with the generated output. This procedure builds upon our previous work [14] and is designed to produce a

simple “yes” for matching answers or “no” otherwise. An individual’s fitness in each generation is computed as the average matching across the reference evaluation dataset, which is kept separate from the data samples used in initialization. Beyond quality metrics, the fitness function may further take into account various aspects such as the length of prompts and responses, their associated costs, and other meaningful measures, thereby significantly influencing the GA’s selection of individuals and tailoring the solution to specific domain requirements.

Datasets: To validate our method on diverse tasks, we conducted experiments on four publicly available datasets: 1) Academic Topic Classification [3], 2) Financial Math Reasoning [5], 3) Bulgarian Academic Q&A [9], and 4) ScienceQA [19]. Although some of these datasets have already been adopted as benchmarks in the NLP community, others are less commonly used. We preprocessed datasets by first randomly isolating an evaluation subset (fixed at 500 rows) from the samples used for initialization. To ensure a consistent format and simplify the shareable version of our implementation, we filter all datasets to retain only essential input-output pairs.

Large Language Models: In our study, we employed three distinct LLMs to evaluate prompt optimization using our custom GA: OpenAI’s GPT-4o-mini [17], Mistral NeMo 12B [2], and Meta’s Llama-3.2-11B-Vision-Instruct [1]. We selected these models because they represent a range of different sizes and include both proprietary and open-source options, all while maintaining low costs. Although this set does not reflect the full scope of existing LLMs, it presents unique challenges for prompt engineering, notably in achieving sufficient task contextualization and specificity across various strategies.

4 Experiments and Results

To assess the efficacy of our customized GA in automating prompt engineering across various tasks and models, we tested our GA-driven prompt optimization using the three LLMs and four datasets presented in Section 3. We further enlisted seasoned prompt engineers to work on the same problems with the same models. These practitioners, skilled in utilizing LLMs for a number of computational tasks, had no prior familiarity with the specific problems selected for this study. They were instructed to employ sample data for contextual understanding and problem framing, mirroring the GA’s initialization procedure. Although they were proficient in English, they lacked proficiency in Bulgarian, which was relevant to one of the datasets. Each engineer was allotted four hours per problem, inclusive of computing time. To establish a control baseline, prompts containing only the input data were used to test the models, with no additional instructions. This “no-prompt approach” was designed to isolate the effect of prompt engineering and to indicate whether the datasets had been incorporated into the model’s prior training or fine-tuning.

Fig.1 depicts line plots for each tested LLM: GPT-4o mini, Mistral NeMo, and Llama3.211B. Subfigures (a), (b), (c), and (d) show the comparison among our GA, manual prompt engineering, and the no-prompt baseline for each dataset. Although the inherent strengths and weaknesses of each model, which influence their overall fitness, are not the primary focus here, the findings clearly indicate that our LLM-based GA successfully generated and refined high-quality prompts across all datasets and models. The curves

representing the top individuals demonstrate that nearly every model improved the initial solutions’ fitness after applying our genetic operators. Furthermore, the average curves confirm that the broader population also benefited from the evolutionary process, as expected.

From a quantitative standpoint, direct comparisons of our LLM-based GA with both the no-prompt and manual prompt engineering baselines revealed consistently superior results across all datasets. The overall improvements for these two baselines averaged 25.6% and 21.0%, respectively. While the former illustrates how much our approach builds upon what the LLM can achieve independently, the latter underscores the practicality of a fully automated prompt engineering workflow. These outcomes closely align with the findings of Guo et al. [7] and Chen et al. [4], who documented improvements of up to 25% and 22% over manually crafted prompts, respectively.

From a qualitative perspective, GA-guided LLMs effectively generated diverse prompts using established prompt engineering strategies, including contextual information, role-playing, and step-by-step guidance. Notably, the GA also enabled LLMs to explore multiple strategies in parallel. Specifically, Mistral NeMo addressed the topic categorization task from three distinct angles: inference (open categories), categorization (predefined categories), and interactive learning experience design (comprehensive descriptions of core topics and learning objectives). Furthermore, the LLMs demonstrated the capability to produce prompts entirely in Bulgarian for the Academic Q&A dataset, surpassing the capabilities of prompt engineers.

Finally, it is important to recognize that the effectiveness of LLMs in judging their own outputs varied substantially. While GPT-4o mini and Mistral NeMo tended to be both accurate and consistent in their assessments, Meta’s Llama3.2 11B displayed more erratic performance, frequently overestimating the equivalence of dissimilar responses. Nonetheless, one can adjust the meta prompts to better fit specific needs and address any subjectivity that may arise in these quality evaluations.

5 Conclusion

The transition to fully automated prompt engineering represents a significant advancement in LLMs’ practical utilization and general adoption. This study underscores how GA-optimized prompts can accelerate the deployment of LLMs by specialized industries, enabling these models to address domain-specific problems with little to no human effort. Our customized GA approach successfully addresses the limitations of manual prompt engineering, as well as the constraints of existing automated and semi-automated methods. Our findings can be organized into three main contributions: first, they demonstrate that GA-guided LLMs systematically take on tasks commonly managed by human specialists; second, they confirm that certain LLMs can reliably judge their own outputs and, thus, guide optimization through the fitness function; and lastly, they provide clear evidence that GA-guided LLMs can surpass manual prompt engineering in carrying out domain-specific tasks, consistent with insights reported in related research. Ultimately, we anticipate that as LLMs continue to evolve, methodologies similar to the one presented here will gain even greater attention, propelling advancements in specialized applications and reshaping the ways in which humans interact with these models.

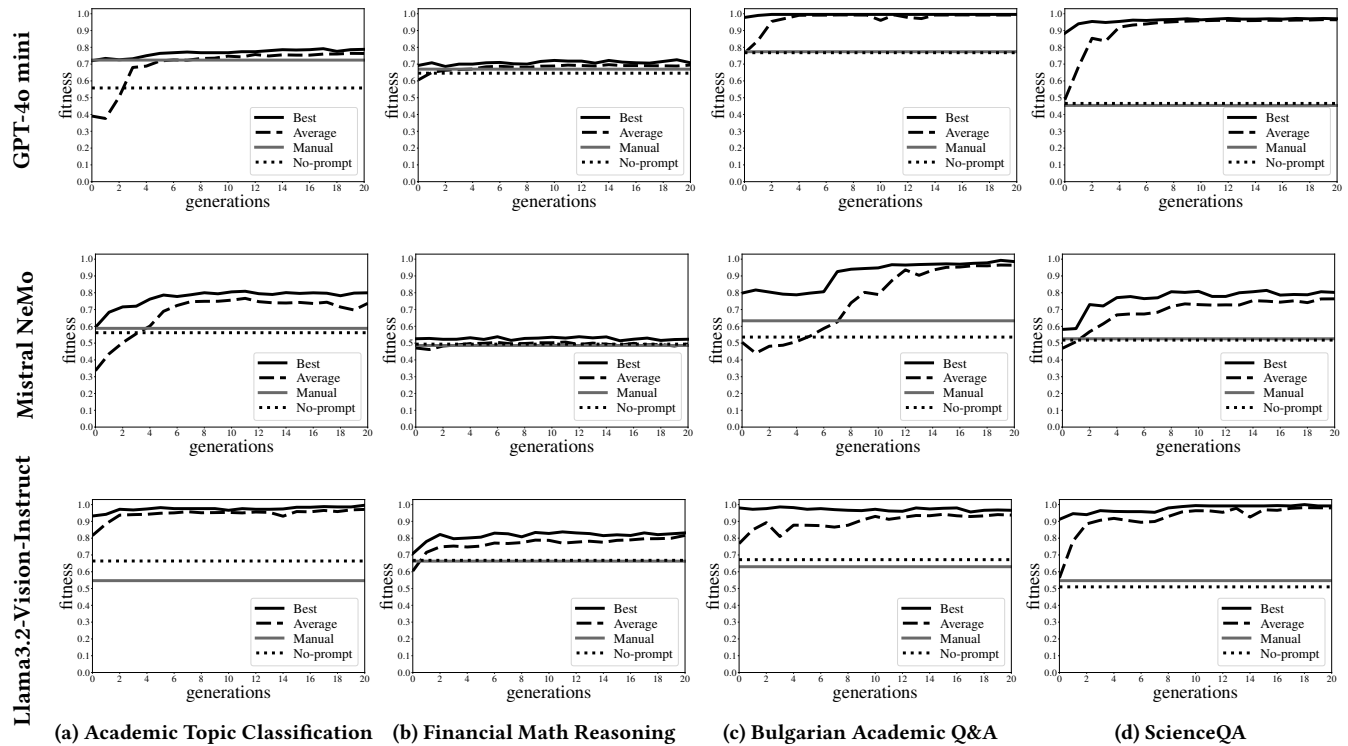


Figure 1: Plots comparing the Best fitness per generation, the Average fitness per generation, the performance achieved by Manual efforts, and the performance achieved by the No-prompt baseline across all tested LLMs and datasets.

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